Ravindra Chowdary Jonnagadla

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Education

SRM Institute of Science and Technology Chennai, India
B.Tech in Computer Science and Engineering – Cloud Computing (CGPA: 8.01/10) May 2026

Technical Skills

Languages: Python, C++, SQL, PostgreSQL, MySQL, JavaScript, TypeScript, Bash, Shell Scripting
AI & Machine Learning: PyTorch, TensorFlow, Scikit-learn, Hugging Face, Whisper, WGAN-GP, PPO Reinforcement Learning, NLP
Cloud & DevOps: AWS, Docker, Kubernetes, Terraform, CI/CD, GitHub Actions, Linux, Git
Backend & Web: FastAPI, Flask, Django, Node.js, Next.js, React, RESTful APIs, Microservices, Redis, Celery
Certifications: AWS Certified Cloud Practitioner – Credly Verified Credential (Badge ID: 3dc45737)

Professional Experience

Waani – Conversational Voice AI Platform Chennai, India
Lead Engineer & Co-Founder July 2026 – Present

- Architected an end-to-end voice agent integrating telephony protocols, streaming STT (Sarvam AI), and LLM inference, achieving $<$800ms p95 voice-to-voice turn-around time across 5,000+ simulated pilot calls.
- Engineered conversational state machines and dynamic intent branching for automated lead qualification and appointment scheduling, configuring automated fallback logic for human operator handoff.
- Engineered an asynchronous dual-channel ring buffer with edge Voice Activity Detection (VAD) over WebSockets to support instant barge-in and mitigate jitter-induced buffer underruns.

LynkZen – Instagram DM Automation SaaS Chennai, India
Lead Engineer & Co-Founder April 2026 – June 2026

- Built and launched an Instagram automation SaaS using Next.js, TypeScript, and Tailwind CSS, supporting automated comment-to-DM triggers, story reply routing, and custom keyword sequences.
- Developed an interactive node-based visual workflow builder allowing creators to configure multi-step conditional messaging funnels and automated response paths without writing code.
- Onboarded **3,000 active beta users for closed private testing**, engineering sliding-window rate-limiting via atomic Redis Lua scripts to ensure 100% compliance with Meta Graph API quotas.

Selected Technical Projects

AMERS: Multimodal Emotion Recognition System | *PyTorch, Signal Processing* IEEE ICETEA 2026

- Developed a multimodal emotion recognition pipeline combining raw EEG signals and speech features, implementing Cross-Modal Mutual Attention (CMMA) to capture inter-modal correlations.
- Integrated Emotion-Aware Gating (EAG) with WGAN-GP for synthetic EEG augmentation, applying Proximal Policy Optimization (PPO) to dynamically calibrate synthetic-to-real sampling ratios against discriminator reward.
- Evaluated on DEAP and SEED multimodal benchmarks across 4 emotion classes, achieving **82.55% accuracy** (+7.4% over unimodal baselines), **0.83 macro F1-score**, and **0.77 Cohen's kappa**.

SmartCap: Real-Time Subtitle Generation | *Python, Whisper, FFmpeg* IEEE ICICV 2025

- Built an automated audio-visual transcription tool using OpenAI Whisper, implementing word-level acoustic timestamp alignment to generate synchronized video subtitles (.srt/.vtt).
- Engineered an asynchronous media worker pipeline using FFmpeg stream filters to isolate acoustic tracks, normalize dynamic audio levels, and burn formatted subtitles onto video.

Smart QR-Based University Parking System | *Flask, Python, Twilio API* IEEE CONIT 2025

- Developed a campus parking management application with HMAC-SHA256 signed dynamic QR code generation, timestamped nonces, and Twilio OTP user authentication.
- Implemented a Progressive Web App (PWA) with real-time slot occupancy tracking and ACID row-level database locking, achieving a **98.7% booking transaction success rate** during peak traffic simulations.

Publications

- **IEEE ICETEA 2026:** "Adaptive GAN-Augmented EEG–Speech Emotion Recognition via Deep Reinforcement Learning" – Int. Conf. on Emerging Technologies in Engineering Applications, Co-Author.
- **IEEE ICICV 2025:** "SmartCap: Real-Time Synchronized Subtitles for Videos" – Published in IEEE Xplore (Doc ID: 11085491), Sole/Lead Author.
- **IEEE CONIT 2025:** "Smart QR-Based University Parking Management System" – Published in IEEE Xplore (Doc ID: 11166870), Co-Author.